

follows as many as 3% of sporadic cases. About one-fifth of secondary cases are actually co-primary cases—i.e., cases that occur soon after the primary case and in which transmission is presumed to have originated from the same third party. The rate of secondary cases is highest during the week after presentation of the index case. The risk falls rapidly but remains above baseline for up to 1 year after the index case; 30% of secondary cases occur in the first week, 20% in the second week, and most of the remainder over the next 6 weeks. In outbreaks of meningococcal disease, mass prophylaxis has been used; however, limited data support population intervention, and significant concerns have arisen about adverse events and the development of resistance. For these reasons, prophylaxis is usually restricted to (1) persons at greatest risk who are intimate and/or household contacts of the index case and (2) health care workers who have been directly exposed to respiratory secretions. In most cases, members of wider communities (e.g., at schools or colleges) are not offered prophylaxis.

The aim of prophylaxis is to eradicate colonization of close contacts with the strain that has caused invasive disease in the index case. Prophylaxis should be given to all contacts at the same time to avoid recolonization by meningococci transmitted from untreated contacts and should also be used as soon as possible to treat early disease in secondary cases. If the index patient is treated with an antibiotic that does not reliably clear colonization (e.g., penicillin), he or she should be given a prophylactic agent at the end of treatment to prevent relapse or onward transmission. Although rifampin has been most widely used and studied, it is not the optimal agent because it fails to eradicate carriage in 15–20% of cases, rates of adverse events have been high, compliance is affected by the need for four doses, and emerging resistance has been reported. Ceftriaxone as a single IM or IV injection is highly (97%) effective in carriage eradication and can be used at all ages and in pregnancy. Reduced susceptibility of isolates to ceftriaxone has occasionally been reported. Ciprofloxacin or ofloxacin is preferred in some countries; these agents are highly effective and can be administered by mouth but are not recommended in pregnancy. Resistance to fluoroquinolones has been reported in some meningococci in North America, Europe, and Asia.

In documented capsular group A, B, C, Y, or W disease, contacts may be offered immunization (with either the MenACWY conjugate vaccine or the MenB vaccine, as appropriate) in addition to chemoprophylaxis to provide protection beyond the duration of antibiotic therapy. Mass vaccination has been used successfully to control disease during outbreaks in closed communities (educational and military establishments) as well as during epidemics in open communities.

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Gonococcal Infections

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DEFINITION

Gonorrhea is a sexually transmitted infection (STI) of epithelium and commonly manifests as cervicitis, urethritis, proctitis, and conjunctivitis. If untreated, infections at these sites can lead to local complications such as endometritis, salpingitis, tuboovarian abscess, Bartholinitis, peritonitis, and perihepatitis in female patients; periurethritis and epididymitis in male patients; and ophthalmia neonatorum in newborns. Disseminated gonococemia is an uncommon event whose manifestations include skin lesions, tenosynovitis, arthritis, and (in rare cases) endocarditis or meningitis.

MICROBIOLOGY

Neisseria gonorrhoeae is a gram-negative, nonmotile, non-spore-forming organism that grows singly and in pairs (i.e., as monococci and diplococci, respectively). Exclusively a human pathogen, the gonococcus contains, on average, three genome copies per coccal unit; this polyploidy permits a high level of antigenic variation and the survival of the organism in its host. Gonococci, like all other *Neisseria* species, are oxidase positive. They are distinguished from other neisseriae by their ability to grow on selective media and to use glucose but not maltose, sucrose, or lactose.

EPIDEMIOLOGY

The incidence of gonorrhea had been declining steadily in the United States, but in 2018, there were ~616,000 newly reported cases—up 56% since 2015. With 87 million cases estimated by the World Health Organization to have occurred globally in 2016, gonorrhea remains a major public health problem worldwide, is a significant cause of morbidity in developing countries, and may play a role in enhancing transmission of HIV.

Gonorrhea predominantly affects young, nonwhite, unmarried, less educated members of urban populations. The number of reported cases probably represents half of the true number of cases—a discrepancy resulting from underreporting, self-treatment, nonspecific treatment without a laboratory-proven diagnosis, and asymptomatic infection. The number of reported new cases of gonorrhea in the United States rose from ~250,000 in the early 1960s to a high of 1.01 million in 1978. The recorded incidence of gonorrhea in modern times peaked in 1975, with 468 reported new cases per 100,000 population in the United States. This peak was attributable to the interaction of several variables, including improved accuracy of diagnosis, changes in patterns of contraceptive use, and changes in sexual behavior. A decline in the overall incidence of gonorrhea in the United States over the past quarter-century may have reflected increased condom use resulting from public health efforts to curtail HIV transmission. Nevertheless, in 2019, 187.8 new cases per 100,000 population were reported in this country, representing a 1-year increase of 5% and a 92% increase since the historic low in 2009; this figure is the highest among industrialized countries. Simultaneously, antibiotic resistance is increasing in the United States and other countries, prompting the U.S. Centers for Disease Control and Prevention (CDC) to name antibiotic-resistant *N. gonorrhoeae* as one of the three most urgent threats of its kind. At present, the attack rate in the United States is highest among 15- to 24-year-old women and 20- to 29-year-old men; >70% of all reported cases occur in these two groups. From the standpoint of ethnicity, rates are highest among African Americans and lowest among persons of Asian descent.

The incidence of gonorrhea is higher in developing countries than in industrialized nations. The exact incidence of any STI is difficult to ascertain in developing countries because of limited surveillance and variable diagnostic criteria. Extremely high rates of gonorrhea have been reported among aboriginal populations in Namibia and Australia. Studies in Africa have clearly demonstrated that nonulcerative STIs

such as gonorrhea (in addition to ulcerative STIs) are an independent risk factor for the transmission of HIV (Chap. 202).

Gonorrhea is transmitted from males to females more efficiently than in the opposite direction. The rate of transmission to a woman during a single unprotected sexual encounter with an infected man is ~50–70%. Oropharyngeal gonorrhea occurs in ~20% of women who practice fellatio with infected partners. Transmission in either direction by cunnilingus is rare.

In any population, there exists a small minority of individuals who have high rates of new-partner acquisition. These “core-group members” or “high-frequency transmitters” are vital in sustaining STI transmission at the population level. Another instrumental factor in sustaining gonorrhea in the population is the large number of infected individuals who are asymptomatic or have minor symptoms that are ignored. These persons, unlike symptomatic individuals, may not cease sexual activity and therefore may continue to transmit the infection. This situation underscores the importance of contact tracing and empirical treatment of the sex partners of index cases.

■ PATHOGENESIS, IMMUNOLOGY, AND ANTIMICROBIAL RESISTANCE

Outer-Membrane Proteins • **PILI** Fresh clinical isolates of *N. gonorrhoeae* initially form piliated (fimbriated) colonies distinguishable on translucent agar. Pilus expression is rapidly switched off with unselected subculture because of rearrangements in pilus genes. This change is a basis for antigenic variation of gonococci. Piliated strains adhere better to cells derived from human mucosal surfaces and are more virulent in organ culture models and human inoculation experiments than nonpiliated variants. In a fallopian tube explant model, pili mediate gonococcal attachment to nonciliated columnar epithelial cells. This event initiates gonococcal adherence, invasion and transport through these cells to intercellular spaces near the basement membrane or directly into the subepithelial tissue. Pili are also essential for genetic competence and transformation of *N. gonorrhoeae*, which permit horizontal transfer of genetic material between different gonococcal lineages in vivo.

OPACITY-ASSOCIATED PROTEIN Another gonococcal surface protein that is important in adherence to epithelial cells is opacity-associated protein (Opa; formerly called protein II). Opa contributes to intergonococcal adhesion, which is responsible for the opaque nature of gonococcal colonies on translucent agar and the organism’s adherence to a variety of eukaryotic cells, including polymorphonuclear leukocytes (PMNs). Certain Opa variants promote invasion of epithelial cells, and this effect has been linked with the ability of Opa to bind vitronectin, heparan sulfate proteoglycans, and several members of the carcinoembryonic antigen-related cell adhesion molecule (CEACAM) receptor family. Epithelial CEACAM-binding gonococci prevent exfoliation of epithelium through a mechanism that involves nitric oxide that is produced during anaerobic bacterial metabolism and upregulation of CD105 (a member of the transforming growth factor- β receptor family), which may interfere with bacterial clearance. *N. gonorrhoeae* Opa proteins that bind CEACAM1, which is expressed by primary CD4⁺ T lymphocytes, suppress the activation and proliferation of these lymphocytes. Select Opa proteins can engage CEACAM3, which is expressed on neutrophils, with consequent nonopsonic phagocytosis (i.e., phagocytosis independent of antibody and complement) and killing of bacteria.

PORIN Porin (previously designated protein I) is the most abundant gonococcal surface protein. Porin molecules exist as trimers that provide anion-transporting aqueous channels through the otherwise hydrophobic outer membrane. Porin exhibits stable interstrain antigenic variation and forms the basis for gonococcal serotyping. Two main serotypes have been identified; PorB.1A strains are often associated with disseminated gonococcal infection (DGI), whereas PorB.1B strains usually cause local genital infections only. DGI strains are generally resistant to the killing action of normal human serum and do not incite a significant local inflammatory response; therefore, they may not cause symptoms at genital sites. These characteristics may be related to the ability of PorB.1A strains to bind to

complement-inhibitory molecules, resulting in a diminished inflammatory response. Porin can translocate to the cytoplasmic membrane of host cells—a process that could initiate gonococcal endocytosis and invasion. PorB.1B present in outer membrane vesicles shed during bacterial growth inhibits the ability of dendritic cells to induce T-cell proliferation and may contribute to the ability of gonococci to subvert adaptive immunity.

OTHER OUTER-MEMBRANE PROTEINS Other notable outer-membrane proteins include H.8, a lipoprotein that is present in high concentration on the surface of all gonococcal strains and is an excellent target for antibody-based diagnostic testing. Transferrin-binding proteins (Tbp1 and Tbp2), lactoferrin-binding proteins (LbpA and LbpB), and hemoglobin/haptoglobin binding proteins (HpuA and HpuB) are required for scavenging iron from transferrin, lactoferrin, and heme in vivo. Transferrin and iron have been shown to enhance the attachment of iron-deprived *N. gonorrhoeae* to human endometrial cells. TdfH and TdfJ enable gonococci to scavenge host zinc from calprotectin and S100 calcium binding protein A7 (psoriasin). IgA1 protease is produced by *N. gonorrhoeae* and may protect the organism from the action of mucosal IgA.

Lipooligosaccharide Gonococcal lipooligosaccharide (LOS) consists of a lipid A and a core oligosaccharide that lacks the repeating O-carbohydrate antigenic side chain seen in many other gram-negative bacteria. Gonococcal LOS possesses marked endotoxic activity and contributes to the local cytotoxic effect in a fallopian tube model. LOS core sugars undergo a high degree of phase variation under different conditions of growth; this variation reflects genetic regulation and expression of glycotransferase genes that dictate the carbohydrate structure of LOS. These phenotypic changes may affect interactions of *N. gonorrhoeae* with elements of the humoral immune system (antibodies and complement) and may also influence direct binding of organisms to both professional phagocytes and nonprofessional phagocytes (epithelial cells). For example, gonococci that are sialylated at their LOS sites inhibit the classic pathway of complement by reducing binding of IgG and also bind complement factor H to inhibit the alternative pathway of complement. LOS sialylation may also decrease nonopsonic Opa-mediated association with neutrophils and inhibit the oxidative burst in PMNs. The binding of the unsialylated terminal lactosamine residue of LOS to an asialoglycoprotein receptor on male epithelial cells facilitates adherence and subsequent gonococcal invasion of these cells. Moreover, oligosaccharide structures in LOS can modulate host immune responses. For example, the terminal monosaccharide expressed by LOS determines the C-type lectin receptor on dendritic cells that is targeted by the bacteria. In turn, the specific C-type lectin receptor engaged influences whether a T_H1- or T_H2-type response is elicited; the latter response may be less favorable for clearance of gonococcal infection.

Host Factors In addition to gonococcal structures that interact with epithelial cells, host factors seem to be important in mediating entry of gonococci into nonphagocytic cells. Activation of phosphatidylcholine-specific phospholipase C and acidic sphingomyelinase by *N. gonorrhoeae*, which results in the release of diacylglycerol and ceramide, is a requirement for the entry of *N. gonorrhoeae* into epithelial cells. Ceramide accumulation within cells leads to apoptosis, which may disrupt epithelial integrity and facilitate entry of gonococci into subepithelial tissue. Release of chemotactic factors as a result of complement activation contributes to inflammation, as does the toxic effect of LOS in provoking the release of inflammatory cytokines.

The importance of humoral immunity in host defenses against neisserial infections is best illustrated by the predisposition of persons deficient in terminal complement components (C5 through C9) to have recurrent bacteremic gonococcal infections and recurrent meningococcal meningitis or meningococcemia. Gonococcal porin induces T cell-proliferative responses in persons with urogenital gonococcal disease. A significant increase in porin-specific interleukin (IL) 4-producing CD4⁺ as well as CD8⁺ T lymphocytes is seen in individuals with mucosal gonococcal disease. A portion of these lymphocytes

that show a porin-specific T_H2 -type response could traffic to mucosal surfaces and play a role in immune protection against the disease. Few data clearly indicate that protective immunity is acquired from a previous gonococcal infection, although bactericidal and opsonophagocytic antibodies to porin and LOS may offer partial protection. On the other hand, women who are infected and acquire high levels of antibody to another outer-membrane protein, Rmp (reduction modifiable protein, formerly called protein III), may be especially likely to become reinfected with *N. gonorrhoeae* because Rmp antibodies block the effect of bactericidal antibodies to porin and LOS. Rmp shows little, if any, interstrain antigenic variation; therefore, Rmp antibodies potentially may block antibody-mediated killing of all gonococci. The mechanism of blocking has not been fully characterized, but Rmp antibodies may noncompetitively inhibit binding of porin and LOS antibodies because of the proximity of these structures in the gonococcal outer membrane. In male volunteers who have no history of gonorrhea, the net effect of these events may influence the outcome of experimental challenge with *N. gonorrhoeae*. Because Rmp bears extensive homology to enterobacterial OmpA and meningococcal class 4 proteins, it is possible that these blocking antibodies result from prior exposure to cross-reacting proteins from these species and also play a role in first-time infection with *N. gonorrhoeae*.

Gonococcal Resistance to Antimicrobial Agents It is no surprise that *N. gonorrhoeae*, with its remarkable capacity to alter its antigenic structure and adapt to changes in the microenvironment, has become resistant to numerous antibiotics. The first effective agents against gonorrhea were the sulfonamides, which were introduced in the 1930s and became ineffective within a decade. Penicillin was then used as the drug of choice for the treatment of gonorrhea. By 1965, 42% of gonococcal isolates had developed low-level resistance to penicillin G. Resistance due to the production of penicillinase arose later.

Gonococci become fully resistant to antibiotics either by chromosomal mutations or by acquisition of R factors (plasmids). Two types of chromosomal mutations have been described. The first type, which is drug specific, is a single-step mutation leading to high-level resistance. The second type involves mutations at several chromosomal loci that combine to determine the level as well as the pattern of resistance. Strains with mutations in chromosomal genes were first observed in the late 1950s. As recently as 2007, chromosomal mutations accounted for resistance to penicillin, tetracycline, or both in ~16% of strains surveyed in the United States.

β -Lactamase (penicillinase)-producing strains of *N. gonorrhoeae* (PPNG) carrying β -lactamase plasmids had rapidly spread worldwide by the early 1980s. *N. gonorrhoeae* strains with plasmid-borne tetracycline resistance (TRNG) can mobilize some β -lactamase plasmids, and PPNG and TRNG occur together, sometimes along with strains exhibiting chromosomally mediated resistance (CMRNG). Penicillin, ampicillin, and tetracycline are no longer reliable for the treatment of gonorrhea and should not be used.

Quinolone-containing regimens also were recommended for treatment of gonococcal infections; the fluoroquinolones offered the advantage of antichlamydial activity when administered for 7 days. However, quinolone-resistant *N. gonorrhoeae* (QRNG) appeared soon after these agents were first used to treat gonorrhea. QRNG is particularly common in the Pacific Islands (including Hawaii) and Asia, where, in certain areas, all gonococcal strains are now resistant to quinolones. At present, QRNG is also common in parts of Europe and the Middle East. In the United States, QRNG has been identified in all areas but predominantly in states on the Pacific coast, where resistant strains were first seen. Alterations in DNA gyrase and topoisomerase IV have been implicated as mechanisms of fluoroquinolone resistance.

Resistance to spectinomycin, which has been used in the past as an alternative agent, has been reported. Because this agent usually is not associated with resistance to other antibiotics, spectinomycin can be reserved for use against multidrug-resistant strains of *N. gonorrhoeae*. Nevertheless, outbreaks caused by strains resistant to spectinomycin have been documented in Korea and England when the drug has been used for primary treatment of gonorrhea.

Third-generation cephalosporins have remained highly effective as single-dose therapy for gonorrhea, but the recent isolation of strains highly resistant to ceftriaxone (minimal inhibitory concentrations [MICs], 2 $\mu\text{g/mL}$) in Japan and some European countries is cause for concern. Even though the MICs of ceftriaxone against certain strains may reach 0.015–0.125 $\mu\text{g/mL}$ (higher than the MICs of 0.0001–0.008 $\mu\text{g/mL}$ for fully susceptible strains), these levels are greatly exceeded in the blood, the urethra, and the cervix when the routinely recommended parenteral dose of ceftriaxone is administered. The rising MICs of oral cefixime (the previously recommended alternative oral third-generation cephalosporin) against *N. gonorrhoeae*, combined with this drug's limited capacity to reach levels sufficiently higher than MICs in the blood, the urethra, the cervix, and especially the pharynx, have resulted in the removal of cefixime from the list of first-line agents for treatment of uncomplicated gonorrhea. *N. gonorrhoeae* strains with reduced susceptibility to ceftriaxone and cefixime (i.e., cephalosporin-intermediate/resistant strains) contain mutations in (1) the *penA* allele, which is the principal resistance determinant and encodes a penicillin-binding protein (PBP2) whose sequence can differ in up to 60–70 amino acids from that of wild-type PBP2; (2) the multiple transferable resistance regulator (*mtrR*) gene that results in increased drug efflux through the MtrCDE efflux pump; and (3) *penB*, which decreases drug influx through PorB.

Resistance to azithromycin can result from alterations of the ribosomal binding target by azithromycin and—as with cephalosporins—the over- and underexpression of efflux and influx systems. Combined resistance to cephalosporins and azithromycin has been reported in several instances throughout the world.

CLINICAL MANIFESTATIONS

Gonococcal Infections in Men Acute urethritis is the most common clinical manifestation of gonorrhea in male patients. The usual incubation period after exposure is 2–7 days, although the interval can be longer and most men remain asymptomatic. Strains of the PorB.1A serotype tend to cause a greater proportion of cases of mild and asymptomatic urethritis than do PorB.1B strains. When they occur, urethral discharge and dysuria, usually without urinary frequency or urgency, are the major symptoms. The discharge initially is scant and mucoid but becomes profuse and purulent within a day or two. Gram's staining of the urethral discharge may reveal PMNs and gram-negative intracellular monococci and diplococci (Fig. 156-1). The clinical manifestations of gonococcal urethritis are usually more severe and overt than those of nongonococcal urethritis, including urethritis caused by *Chlamydia trachomatis* (Chap. 189); however, exceptions are common, and it is often impossible to differentiate the causes of urethritis on clinical grounds alone. The majority of cases of urethritis seen in the United States today are not caused by *N. gonorrhoeae* and/or *C. trachomatis*. Although a number of other organisms may be responsible, many cases do not have a specific etiologic agent identified. Certain clones of *Neisseria meningitidis*, the second member of the pathogenic

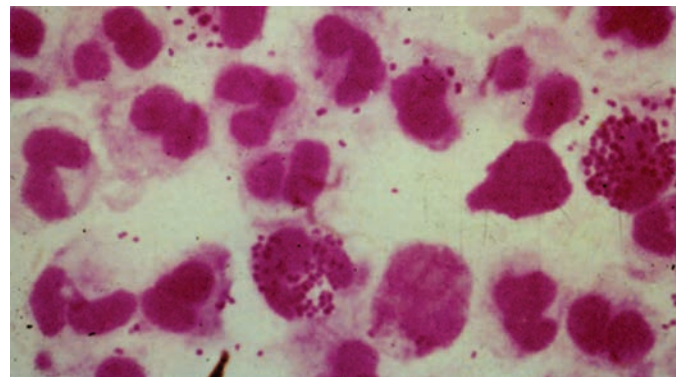


FIGURE 156-1 Gram's stain of urethral discharge from a male patient with gonorrhea shows gram-negative intracellular monococci and diplococci. (Source: © All rights reserved. Canadian Guidelines on Sexually Transmitted Infections. Public Health Agency of Canada, modified 2020. Adapted and reproduced with permission from the Minister of Health, 2021.)

Neisseria species, have been associated with urethritis in men who have sex with men (MSM) in Europe and in heterosexual men in the southern and midwestern United States.

Most symptomatic men with gonorrhea seek treatment and cease to be infectious. The remaining men, who are largely asymptomatic, accumulate in number over time and constitute about two-thirds of all infected men at any point in time; together with men incubating the organism who shed the organism but are asymptomatic, they serve as the source of spread of infection. Before the antibiotic era, symptoms of urethritis persisted for ~8 weeks. Epididymitis is now an uncommon complication, and gonococcal prostatitis occurs rarely, if at all. Other unusual local complications of gonococcal urethritis include edema of the penis due to dorsal lymphangitis or thrombophlebitis, submucous inflammatory “soft” infiltration of the urethral wall, periurethral abscess or fistula, inflammation or abscess of Cowper’s gland, and seminal vesiculitis. Balanitis may develop in uncircumcised men.

Gonococcal Infections in Women • GONOCOCCAL CERVICITIS

Mucopurulent cervicitis is a common STI diagnosis in American women and may be caused by *N. gonorrhoeae*, *C. trachomatis*, and other organisms, including *Mycoplasma genitalium* (Chap. 188). Cervicitis may coexist with candidal or trichomonal vaginitis. *N. gonorrhoeae* primarily infects the columnar epithelium of the cervical os. Bartholin’s glands occasionally become infected.

Women infected with *N. gonorrhoeae* usually develop symptoms. However, women who either remain asymptomatic or have only minor symptoms may delay seeking medical attention. These minor symptoms may include scant vaginal discharge issuing from the inflamed cervix (without vaginitis or vaginosis per se) and dysuria (often without urgency or frequency) that may be associated with gonococcal urethritis. Although the incubation period of gonorrhea is less well defined in women than in men, symptoms usually develop within 10 days of infection and are more acute and intense than those of chlamydial cervicitis.

The physical examination reveals a mucopurulent discharge (mucopus) issuing from the cervical os or a reddened (inflamed) cervix even in the absence of reported symptoms. Because Gram’s stain is not sensitive for the diagnosis of gonorrhea in women, specimens should be submitted for culture or a nonculture assay (see “Laboratory Diagnosis,” below). Edematous and friable cervical ectopy and endocervical bleeding induced by gentle swabbing are more often seen in chlamydial infection. Gonococcal infection may extend deep enough to produce dyspareunia and lower abdominal or back pain. In such cases, it is imperative to consider a diagnosis of pelvic inflammatory disease (PID) and to administer treatment for that disease (Chaps. 136 and 189).

N. gonorrhoeae may also be recovered from the urethra and rectum of women with cervicitis, but these are rarely the only infected sites. Urethritis in women may produce symptoms of internal dysuria, which is often attributed to “cystitis.” Pyuria in the absence of bacteriuria visible on Gram’s stain of unspun urine, accompanied by urine cultures that fail to yield $>10^2$ colonies of bacteria usually associated with urinary tract infection, signifies the possibility of urethritis usually due to *C. trachomatis*. Urethral infection with *N. gonorrhoeae* also may occur in this context, but in this instance, urethral cultures are usually positive.

GONOCOCCAL VAGINITIS The vaginal mucosa of healthy women is lined by stratified squamous epithelium and is rarely infected by *N. gonorrhoeae*. However, gonococcal vaginitis can occur in anerogenic women (e.g., prepubertal girls and postmenopausal women), in whom the vaginal stratified squamous epithelium is often thinned down to the basilar layer, which can be infected by *N. gonorrhoeae*. The intense inflammation of the vagina makes the physical (speculum and bimanual) examination extremely painful. The vaginal mucosa is red and edematous, and an abundant purulent discharge is often present. Infection in the urethra and in Skene’s and Bartholin’s glands often accompanies gonococcal vaginitis. Inflamed cervical erosion or abscesses in nabothian cysts may also occur. Coexisting cervicitis may result in pus in the cervical os.

Anorectal Gonorrhea Because the female anatomy permits the spread of cervical exudate to the rectum, *N. gonorrhoeae* is sometimes recovered from the rectum of women with uncomplicated gonococcal cervicitis. The rectum is the sole site of infection in only 5% of women with gonorrhea. Such women are usually asymptomatic but occasionally have acute proctitis manifested by anorectal pain or pruritus, tenesmus, purulent rectal discharge, and rectal bleeding. Among MSM, the frequency of gonococcal infection, including rectal infection, fell by $\geq 90\%$ throughout the United States in the early 1980s. A resurgence of gonorrhea among MSM has been documented in several cities since the 1990s, the estimated rates of reported cases having more than doubled in a recent 3-year period. Gonococcal isolates from the rectum of MSM tend to be more resistant to antimicrobial agents than are gonococcal isolates from other sites. Gonococcal isolates with a mutation in *mtrR* or in the promoter region of the gene that encodes for this transcriptional regulator develop increased resistance to antimicrobial hydrophobic agents such as bile acids and fatty acids in feces and thus are found with increased frequency in MSM. This situation may have been responsible for higher rates of failure of treatment for rectal gonorrhea with older regimens consisting of penicillin or tetracyclines.

Pharyngeal Gonorrhea Pharyngeal gonorrhea is usually mild or asymptomatic, although symptomatic pharyngitis does occasionally occur with cervical lymphadenitis. The mode of acquisition is oral-genital sexual exposure, with fellatio being a more efficient means of transmission than cunnilingus. In certain female adolescent populations in the United States, pharyngeal gonorrhea has become as common as genital gonorrhea. Most cases resolve spontaneously, and transmission from the pharynx to sexual contacts is rare. Pharyngeal infection almost always coexists with genital infection. Swabs from the pharynx should be plated directly onto gonococcal selective media. Pharyngeal colonization with *N. meningitidis* needs to be differentiated from that with other *Neisseria* species. Because commensal oropharyngeal neisseriae are often resistant to antimicrobials, horizontal gene transfer between these organisms and *N. gonorrhoeae* may be important in the development of antimicrobial resistance of *N. gonorrhoeae*.

Ocular Gonorrhea in Adults Ocular gonorrhea in an adult usually results from autoinoculation of *N. gonorrhoeae* from an infected genital site. As in genital infection, the manifestations range from severe to occasionally mild or asymptomatic disease. The variability in clinical manifestations may be attributable to differences in the ability of the infecting strain to elicit an inflammatory response. Infection may result in a markedly swollen eyelid, severe hyperemia and chemosis, and a profuse purulent discharge. The massively inflamed conjunctiva may be draped over the cornea and limbus. Lytic enzymes from the infiltrating PMNs occasionally cause corneal ulceration and rarely cause perforation.

Prompt recognition and treatment of this condition are of paramount importance. Gram’s stain and culture of the purulent discharge establish the diagnosis. Genital cultures also should be performed.

Gonorrhea in Pregnant Women, Neonates, and Children

Gonorrhea in pregnancy can have serious consequences for both the mother and the infant. Recognition of gonorrhea early in pregnancy also identifies a population at risk for other STIs, particularly chlamydial infection, syphilis, and trichomoniasis. The risks of salpingitis and PID—conditions associated with a high rate of fetal loss—are highest during the first trimester. Pharyngeal infection, most often asymptomatic, may be more common during pregnancy because of altered sexual practices. Prolonged rupture of the membranes, premature delivery, chorioamnionitis, funisitis (infection of the umbilical cord stump), and sepsis in the infant (with *N. gonorrhoeae* detected in the newborn’s gastric aspirate during delivery) are common complications of maternal gonococcal infection at term. Other conditions and microorganisms, including *Mycoplasma hominis*, *Mycoplasma genitalium*, *Ureaplasma urealyticum*, *C. trachomatis*, and bacterial vaginosis (often accompanied by infection with *Trichomonas vaginalis*), have been associated with similar complications.

The most common form of gonorrhea in neonates is **ophthalmia neonatorum**, which results from exposure to infected cervical secretions during parturition. Ocular neonatal instillation of a prophylactic agent (e.g., 1% silver nitrate eye drops or ophthalmic preparations containing erythromycin or tetracycline) prevents ophthalmia neonatorum but is not effective for its treatment, which requires systemic antibiotics. The clinical manifestations are acute and usually begin 2–5 days after birth. An initial nonspecific conjunctivitis with a serosanguineous discharge is followed by tense edema of the eyelids, chemosis, and a profuse, thick, purulent discharge. Corneal ulcerations that result in nebulae or perforation may lead to anterior synechiae, anterior staphyloma, panophthalmitis, and blindness. Infections described at other mucosal sites in infants, including vaginitis, rhinitis, and anorectal infection, are likely to be asymptomatic. Pharyngeal colonization has been demonstrated in 35% of infants with gonococcal ophthalmia, and coughing is the most prominent symptom in these cases. Septic arthritis (see below) is the most common manifestation of systemic infection or DGI in the newborn. The onset usually comes at 3–21 days of age, and polyarticular involvement is common. Sepsis, meningitis, and pneumonia are seen in rare instances.

Any STI in children beyond the neonatal period raises the possibility of sexual abuse. Gonococcal vulvovaginitis is the most common manifestation of gonococcal infection in children beyond infancy. Anorectal and pharyngeal infections are common in these children and are frequently asymptomatic. The urethra, Bartholin's and Skene's glands, and upper genital tract are rarely involved. All children with gonococcal infection should also be evaluated for chlamydial infection, syphilis, and possibly HIV infection.

Gonococcal Arthritis Disseminated gonococcal infection (DGI; gonococcal arthritis) results from gonococcal bacteremia. In the 1970s, DGI occurred in ~0.5–3% of persons with untreated gonococcal mucosal infection. The lower incidence of DGI at present is probably attributable to a decline in the prevalence of particular strains that are likely to disseminate. Nonetheless, sporadic outbreaks of DGI still occur in North America. DGI strains resist the bactericidal action of human serum and generally do not incite inflammation at genital sites, probably because of limited generation of chemotactic factors. Strains recovered from DGI cases in the 1970s were often of the PorB.1A serotype, were highly susceptible to penicillin, and had special growth requirements—including arginine, hypoxanthine, and uracil—that made the organism more fastidious and more difficult to isolate.

Menstruation is a risk factor for dissemination, and approximately two-thirds of cases of DGI are in women. In about half of affected women, symptoms of DGI begin within 7 days of onset of menses. **Complement deficiencies**, especially of the components involved in the assembly of the membrane attack complex (C5 through C9), predispose to neisserial bacteremia, and persons with more than one episode of DGI should be screened with an assay for total hemolytic complement activity. DGI is also associated with the use of the complement C5-blocking monoclonal antibody eculizumab.

The clinical manifestations of DGI have sometimes been classified into two stages: a bacteremic stage, which is less common today, and a joint-localized stage with suppurative arthritis. A clear-cut progression usually is not evident. Patients in the bacteremic stage have higher temperatures, and chills more frequently accompany their fever. Painful

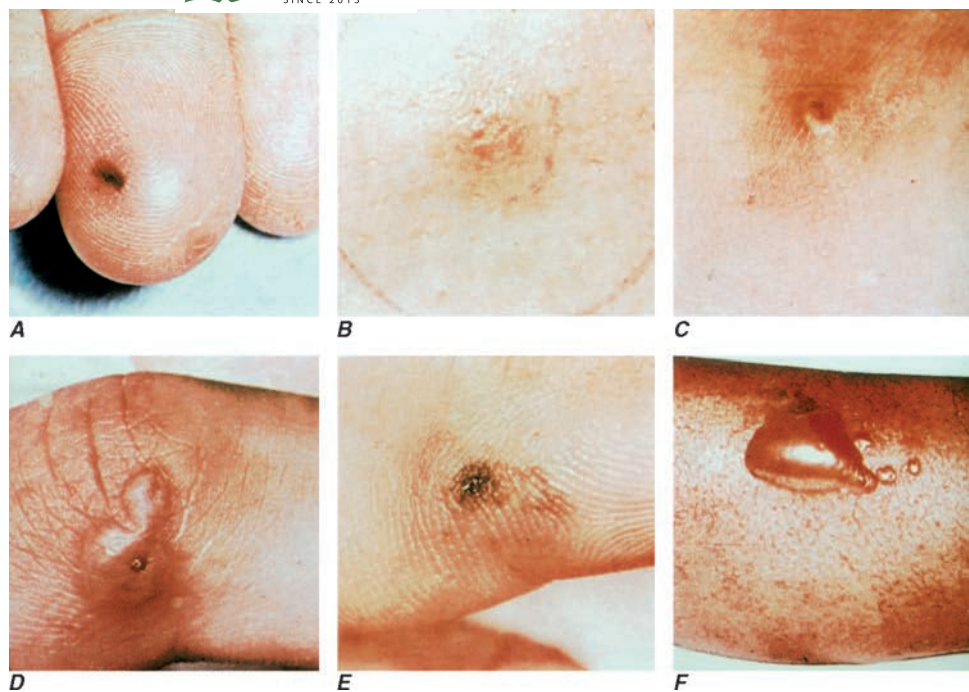


FIGURE 156-2 Characteristic skin lesions in patients with proven gonococcal bacteremia. The lesions are in various stages of evolution. **A.** Very early petechia on finger. **B.** Early papular lesion, 7 mm in diameter, on lower leg. **C.** Pustule with central eschar resulting from early petechial lesion. **D.** Pustular lesion on finger. **E.** Mature lesion with central necrosis (black) on hemorrhagic base. **F.** Bullae on anterior tibial surface. (Reprinted with permission from TF Murphy, GI Parameswaran: *Clin Infect Dis* 49:124, 2009, with permission. © 2009 Infectious Diseases Society of America.)

joints are common and often occur together with tenosynovitis and skin lesions. Polyarthralgias usually include the knees, elbows, and more distal joints; the axial skeleton is generally spared. Skin lesions are seen in ~75% of patients and include papules and pustules, often with a hemorrhagic component (Fig. 156-2; see also Fig. A1-43). Other manifestations of noninfectious dermatitis, such as nodular lesions, urticaria, and erythema multiforme, have been described. These lesions are usually on the extremities and number between 5 and 40. The differential diagnosis of the bacteremic stage of DGI includes reactive arthritis, acute rheumatoid arthritis, sarcoidosis, erythema nodosum, drug-induced arthritis, and viral infections (e.g., hepatitis B and acute HIV infection). The distribution of joint symptoms in reactive arthritis differs from that in DGI (Fig. 156-3), as do the skin and genital manifestations (Chap. 362).

Suppurative arthritis involves one or two joints, most often the knees, wrists, ankles, and elbows (in decreasing order of frequency); other joints occasionally are involved. Most patients who develop gonococcal septic arthritis do so without prior polyarthralgias or skin lesions; in the absence of symptomatic genital infection, this disease cannot be distinguished from septic arthritis caused by other pathogens. The differential diagnosis of acute arthritis in young adults is discussed in Chap. 130. Rarely, osteomyelitis complicates septic arthritis involving small joints of the hand.

Gonococcal endocarditis, although rare today, was a relatively common complication of DGI in the preantibiotic era, accounting for about one-quarter of reported cases of endocarditis. Another unusual complication of DGI is meningitis.

Gonococcal Infections in HIV-Infected Persons The association between gonorrhea and the acquisition of HIV has been demonstrated in several well-controlled studies, mainly in Kenya and Zaire. The nonulcerative STIs enhance the transmission of HIV three- to fivefold; transmission of HIV-infected immune cells and increased viral shedding by persons with urethritis or cervicitis may contribute (Chap. 202). HIV has been detected by polymerase chain reaction (PCR) more commonly in ejaculates from HIV-positive men with gonococcal urethritis than in those from HIV-positive men with nongonococcal urethritis. PCR positivity diminishes

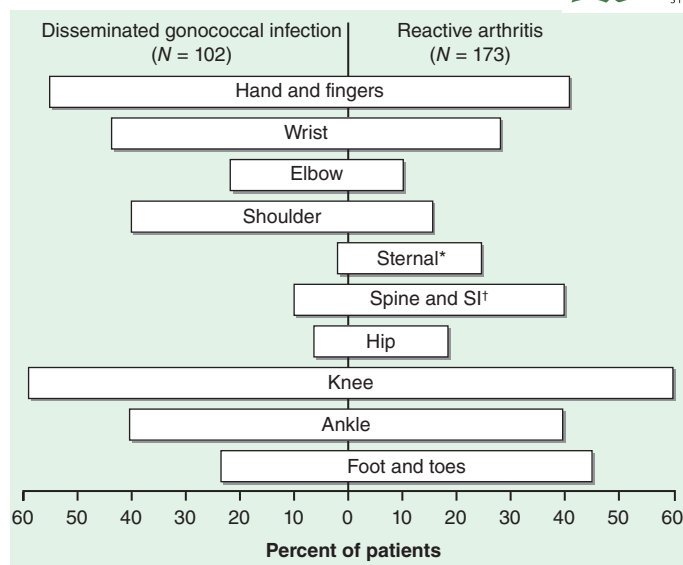


FIGURE 156-3 Distribution of joints with arthritis in 102 patients with disseminated gonococcal infection and 173 patients with reactive arthritis. *Includes the sternoclavicular joints. *SI, sacroiliac joint.

twofold after appropriate therapy for urethritis. Not only does gonorrhea enhance the transmission of HIV, but it may also increase the individual's risk for acquisition of HIV. A proposed mechanism is the significantly greater number of CD4⁺ T lymphocytes and dendritic cells that can be infected by HIV in endocervical secretions from women with nonulcerative STIs than in those from women with ulcerative STIs.

LABORATORY DIAGNOSIS

A rapid diagnosis of gonococcal infection in men may be obtained by Gram's staining of urethral exudates (Fig. 156-1). The detection of gram-negative intracellular monococci and diplococci is usually highly specific and sensitive in diagnosing gonococcal urethritis in symptomatic males but is only ~50% sensitive in diagnosing gonococcal cervicitis. Samples should be collected with Dacron or rayon swabs. Part of the sample should be inoculated onto a plate of modified Thayer-Martin or other gonococcal selective medium for culture. It is important to process all samples immediately because gonococci do not tolerate drying. If plates cannot be incubated immediately, they can be held safely for several hours at room temperature in candle extinction jars prior to incubation. If processing is to occur within 6 h, transport of specimens may be facilitated by the use of nonnutritive swab transport systems such as Stuart or Amies medium. For longer holding periods (e.g., when specimens for culture are to be mailed), culture media with self-contained CO₂-generating systems (such as the JEMBEC or Gono-Pak systems) may be used. Specimens should also be obtained for the diagnosis of chlamydial infection (Chap. 189).

PMNs are often seen in the endocervix on a Gram's stain, and an abnormally increased number (≥30 PMNs per field in five 1000× oil-immersion microscopic fields) establishes the presence of an inflammatory discharge. Unfortunately, the presence or absence of gram-negative intracellular monococci or diplococci in cervical smears does not accurately predict which patients have gonorrhea, and the diagnosis in this setting should be made by culture or another suitable nonculture diagnostic method. The sensitivity of a single endocervical culture is ~80–90%. If a history of rectal sex is elicited, a rectal wall swab (uncontaminated with feces) should be cultured. A presumptive diagnosis of gonorrhea cannot be made on the basis of gram-negative diplococci in smears from the pharynx, where other *Neisseria* species are components of the normal flora.

Several nucleic acid amplification tests (NAATs), including the Roche COBAS AMPLICOR, Gen-Probe Aptima Combo 2, and BD ProbeTec ET, are now widely available on semiautomated or fully automated platforms and are commonly employed diagnostic tests for

gonorrhea. These tests also detect *C. trachomatis* and are more sensitive than culture for identification of either *N. gonorrhoeae* or *C. trachomatis*. The Gen-Probe and BD tests offer the advantage that urine samples can be tested with a sensitivity similar to or greater than that obtained when urethral or cervical swab samples are assessed by other non-NAATs or culture, respectively. A point-of-care NAAT-based test (Binx io) for gonorrhea and chlamydia with a 30-minute turnaround time is now approved by the U.S. Food and Drug Administration (FDA). In MSM, it is important to screen the rectum and pharynx because screening urine alone will miss the majority of cases. A disadvantage of non-culture-based assays is that *N. gonorrhoeae* cannot be grown from the transport systems. Thus, a culture-confirmatory test and formal antimicrobial susceptibility testing, if needed, cannot be performed.

Because of the legal implications, the preferred method for the diagnosis of gonococcal infection in children is a standardized culture. Two positive NAATs, each targeting a different nucleic acid sequence, may be substituted for culture of the cervix or the urethra as legal evidence of infection in children. Although nonculture tests for gonococcal infection have not been approved by the FDA for use with specimens obtained from the pharynx and rectum of infected children, NAATs from these sites are preferred for diagnostic evaluation in adult victims of suspected sexual abuse, especially if the NAATs have been evaluated by the local laboratory and found to be superior. Cultures should be obtained from the pharynx and anus of both girls and boys, the urethra of boys, and the vagina of girls; cervical specimens are not recommended for prepubertal girls. For boys with a urethral discharge, a meatal specimen of the discharge is adequate for culture. Presumptive colonies of *N. gonorrhoeae* should be identified definitively by at least two independent methods.

Blood should be cultured in suspected cases of DGI. The use of Isolator blood culture tubes may enhance the yield. The probability of positive blood cultures decreases after 48 h of illness. Synovial fluid should be inoculated into blood culture broth medium and plated onto chocolate agar rather than selective medium because this fluid is not likely to be contaminated with commensal bacteria. Gonococci are infrequently recovered from early joint effusions containing <20,000 leukocytes/μL but may be recovered from effusions containing >80,000 leukocytes/μL. The organisms are seldom recovered from blood and synovial fluid of the same patient.

TREATMENT

Gonococcal Infections

Treatment failure can lead to continued transmission and the emergence of antibiotic resistance. The importance of adequate treatment with a regimen that the patient will adhere to cannot be overemphasized. Single-dose regimens have been developed for uncomplicated gonococcal infections. Treatment guidelines for gonococcal infections from the Centers for Disease Control and Prevention (CDC) were revised in 2020, and are summarized in Table 156-1. The third-generation cephalosporin ceftriaxone is now recommended as the first-line regimen for use at twice the previous dose (now, 500 mg IM, single dose) based on doubling of mean inhibitory concentrations (MICs) of current strains compared with MICs 20 years ago. The development of decreased sensitivity to ceftriaxone throughout the world will require the development of new effective regimens. Azithromycin, which had been recommended to provide additional treatment of gonorrhea (also to include treatment of chlamydial infection) is no longer recommended as part of a first line regimen. Resistance to azithromycin of U.S. isolates of *N. gonorrhoeae*, which had been less than 0.6% over a number of years, has increased more than sevenfold to 4.6% in the most recent year in which it was reported. If chlamydial infection cannot be excluded, concurrent treatment with doxycycline (100 mg orally twice a day for 7 days) is recommended. The recommendations for uncomplicated gonorrhea apply to HIV-infected as well as HIV-uninfected patients.

TABLE 156-1 Recommended Treatment for Gonococcal Infections:
 Adapted from the 2020 Guidelines for Gonococcal Infection of the
 Centers for Disease Control and Prevention

DIAGNOSIS	TREATMENT OF CHOICE ^a
Uncomplicated gonococcal infection of the cervix, urethra, pharynx ^b , or rectum First-line regimen	Ceftriaxone (500 mg IM, single dose) plus Doxycycline (100 mg orally twice a day for 7 days) for treatment of chlamydial infection if chlamydial infection cannot be excluded
Alternative regimens if ceftriaxone is not available	Gentamicin (240 mg IM, single dose) plus azithromycin (2 g orally as a single dose) ^c Cefixime (800 mg PO, single dose) <i>or</i> spectinomycin (2 g IM, single dose) ^{d,e} plus Doxycycline (100 mg orally twice a day for 7 days) for treatment of chlamydial infection if chlamydial infection cannot be excluded
Epididymitis	See Chap. 136
Pelvic inflammatory disease	See Chap. 136
Gonococcal conjunctivitis in an adult	Ceftriaxone (1 g IM, single dose) ^f
Ophthalmia neonatorum ^g	Ceftriaxone (25–50 mg/kg IV, single dose, not to exceed 125 mg)
Disseminated gonococcal infection ^h Initial therapy ⁱ	Ceftriaxone (1 g IM or IV q24h; recommended) <i>or</i> cefotaxime (1 g IV q8h) <i>or</i> ceftizoxime (1 g IV q8h) Spectinomycin (2 g IM q12h) ^d Cefixime (400 mg PO bid)
Patient tolerant of β -lactam drugs	
Patients allergic to β -lactam drugs	
Continuation therapy ^j	
Meningitis or endocarditis	See text for specific recommendations ^k

^aTrue failure of treatment with a recommended regimen is rare and should prompt an evaluation for reinfection, infection with a drug-resistant strain, or an alternative diagnosis. ^bCeftriaxone is the most reliable agent recommended for treatment of pharyngeal infection. ^c*In vitro* synergistic killing of *N. gonorrhoeae* of gentamicin plus azithromycin is mild to moderate; azithromycin is for treatment of chlamydial infection, primarily. ^dSpectinomycin is unavailable in the United States; in uncomplicated gonococcal infection it should be used at a higher dose (4 g IM, single dose) in areas of the world where increased resistance to spectinomycin exists. ^eSpectinomycin may be ineffective for the treatment of pharyngeal gonorrhea. ^fPlus lavage of the infected eye with saline solution (once). ^gProphylactic regimens are discussed in the text. ^hHospitalization is indicated if the diagnosis is uncertain, if the patient has the joint-localized stage with suppurative arthritis, or if the patient cannot be relied on to adhere to treatment. ⁱAll initial regimens should also include doxycycline (100 mg orally twice a day for 7 days) for treatment of chlamydial infection if chlamydial infection cannot be excluded; gonococcal therapy should be continued for 24–48 h after clinical improvement begins, at which time the switch may be made to an oral agent (e.g., cefixime) if antimicrobial susceptibility can be documented by culture of the causative organism. If no organism is isolated and the diagnosis is secure, then treatment with ceftriaxone should be continued for at least 1 week. ^jHospitalization is indicated to exclude suspected meningitis or endocarditis.

The currently recommended regimen for the treatment of uncomplicated gonococcal infection of the urethra, cervix, rectum, or pharynx (a single IM dose of ceftriaxone) almost always results in an effective cure. Quinolone-containing regimens are no longer recommended in the United States as first-line treatment because of widespread resistance. Rising MICs of cefixime worldwide have led the CDC to discontinue its recommendation of this agent as first-line treatment for uncomplicated gonorrhea. Multicenter trials of treatment for uncomplicated gonorrhea in the United States have shown $\geq 99.5\%$ efficacy of two combination regimens and 96% efficacy in one single-agent regimen: gemifloxacin (320 mg, single oral dose) plus azithromycin (2 g, single oral dose); gentamicin (a single IM dose of 240 mg or, in individuals who weigh ≤ 45 kg, 5 mg/kg) plus azithromycin (2 g, single oral dose), and zoliflodacin (2 or 3 g, single oral dose). At this time, however, none of these regimens

is recommended by CDC as first-line treatment; gentamicin plus azithromycin is recommended as an alternative regimen.

Co-infection with *C. trachomatis* occurs frequently; concurrent treatment with doxycycline (100 mg orally twice daily for 7 days) is effective against chlamydial infection. Spectinomycin has been used as an alternative agent for the treatment of uncomplicated gonococcal infections in penicillin-allergic persons outside the United States but is not currently available in the United States. Of note, the limited effectiveness of spectinomycin for the treatment of pharyngeal infection reduces its utility in populations among whom such infection is common, such as MSM.

Persons with uncomplicated genital or rectal infections who receive ceftriaxone or an alternative regimen do not need a test of cure; however, cultures for *N. gonorrhoeae* should be performed if symptoms persist after therapy with an established regimen, and any gonococci isolated should be tested for antimicrobial susceptibility. Persons with pharyngeal infection should undergo a test of cure regardless of the treatment regimen, 7–14 days after treatment to ensure eradication or detection of a possible treatment failure. Symptomatic gonococcal pharyngitis is more difficult to eradicate than genital infection. Persons who cannot tolerate cephalosporins may be treated with an alternative regimen. Treatment with spectinomycin results in a cure rate of $\leq 52\%$; persons given spectinomycin should have a subsequent pharyngeal sample cultured early (3–5 days) following treatment as a test of cure. A single 2-g dose of azithromycin may be used if the infecting organism is known to be sensitive or in areas where rates of resistance to azithromycin are low. Quinolones may be used if the infecting organism is known to be sensitive. If culture is not readily available and NAAT is positive, every effort should be made to perform a confirmatory culture. All isolates from test-of-cure cultures should undergo antimicrobial susceptibility testing. Because of high rates of reinfection with *N. gonorrhoeae* (and *C. trachomatis*) within 6–12 months, persons previously treated for gonorrhea should be retested 3 months after treatment.

Treatments for gonococcal epididymitis and PID are discussed in Chap. 136. Ocular gonococcal infections in older children and adults should be managed with a single dose of ceftriaxone combined with saline irrigation of the conjunctivae (both undertaken expeditiously), and patients should undergo a careful ophthalmologic evaluation that includes a slit-lamp examination.

DGI, particularly the joint-localized stage with suppurative arthritis, may require higher dosages and longer durations of therapy (Table 156-1). Hospitalization is indicated if the diagnosis is uncertain, if the patient has localized suppurative arthritis that requires aspiration, or if the patient cannot be relied on to comply with treatment. Open drainage is necessary only occasionally—e.g., for management of hip infections that may be difficult to drain percutaneously. Nonsteroidal anti-inflammatory agents may be indicated to alleviate pain and hasten clinical improvement of affected joints.

Gonococcal meningitis and endocarditis should be treated in the hospital with high-dose IV ceftriaxone (1–2 g IV every 12–24 h); therapy should continue for 10–14 days for meningitis and for at least 4 weeks for endocarditis. All persons who experience more than one episode of DGI should be evaluated for complement deficiency.

■ PREVENTION AND CONTROL

Condoms, if properly used, provide effective protection against the transmission and acquisition of gonorrhea as well as other infections that are transmitted to and from genital mucosal surfaces. Spermicidal preparations used with a diaphragm or cervical sponges impregnated with nonoxynol-9 offer some protection against gonorrhea and chlamydial infection. However, the frequent use of preparations that contain nonoxynol-9 is associated with mucosal disruption that paradoxically may enhance the risk of HIV infection in the event of exposure. All patients should be instructed to refer sex partners for evaluation and treatment. All sex partners of persons with gonorrhea

should be evaluated and treated for *N. gonorrhoeae* and *C. trachomatis* infections if their last contact with the patient took place within 60 days before the onset of symptoms or the diagnosis of infection in the patient. If the patient's last potential sexual exposure to infection was >60 days before onset of symptoms or diagnosis, the patient's most recent sex partner should be treated. Partner-delivered medications or prescriptions for medications to treat gonorrhea and chlamydial infection diminish the likelihood of reinfection (or relapse) in the infected patient. In states where it is not prohibited, this approach is an option for partner management. Patients should be instructed to abstain from sexual intercourse until therapy is completed and until they and their sex partners no longer have symptoms. Greater emphasis must be placed on prevention by public health education, individual patient counseling, and behavior modification, particularly the use of condoms. Sexually active persons, especially adolescents, should be offered screening for STIs. For most male patients, NAAT of urine or a urethral swab may be used for screening. Preventing the spread of gonorrhea may help reduce the transmission of HIV. No effective vaccine for gonorrhea is yet available, but efforts to test several candidates are underway including a field trial of a licensed group B meningococcal vaccine (Bexsero), which in prototype form had been shown to reduce the incidence of gonorrhea in a population given the vaccine to control a group B meningococcal epidemic.

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Haemophilus and Moraxella Infections

Timothy F. Murphy

HAEMOPHILUS INFLUENZAE

MICROBIOLOGY

Haemophilus influenzae was first recognized in 1892 by Pfeiffer, who erroneously concluded that the bacterium was the cause of influenza. *H. influenzae* is a small (1- × 0.3-μm) gram-negative organism of variable shape; thus, it is often described as a pleomorphic coccobacillus. In clinical specimens such as cerebrospinal fluid (CSF) and sputum, *H. influenzae* frequently stains only faintly with safranin and therefore can easily be overlooked.

H. influenzae grows both aerobically and anaerobically. Its aerobic growth requires two factors: hemin (X factor) and nicotinamide adenine dinucleotide (V factor). These requirements are used in the clinical laboratory to identify the bacterium. However, using

TABLE 157-1 Characteristics of Type b and Nontypable Strains of *Haemophilus influenzae*

FEATURE	TYPE b STRAINS	NONTYPABLE STRAINS
Capsule	Ribosyl-ribitol phosphate	Unencapsulated
Pathogenesis	Invasive infections due to hematogenous spread	Mucosal infections due to contiguous spread
Clinical manifestations	Meningitis and invasive infections in incompletely immunized infants and children	Otitis media in infants and children; lower respiratory tract infections in adults with chronic bronchitis
Evolutionary history	Basically clonal	Genetically diverse
Vaccine	Highly effective conjugate vaccines	Protein D used as carrier protein in pneumococcal vaccine approved in Europe: GSK Synflorix. Others under development

phenotypic methods for differentiating among *Haemophilus* species has limitations, as the growing number of whole-genome sequences of *Haemophilus* isolates from the human respiratory tract is revealing complex genetic relationships among *Haemophilus* species (see "Diagnosis," below).

Six major serotypes of *H. influenzae* have been identified; designated *a* through *f*, they are based on antigenically distinct polysaccharide capsules. In addition, some strains lack a polysaccharide capsule and are referred to as *nontypable* strains. Type b and nontypable strains are the most relevant strains clinically (Table 157-1), although encapsulated strains other than type b can cause disease. *H. influenzae* was the first free-living organism to have its entire genome sequenced.

The antigenically distinct type b capsule is a linear polymer composed of ribosyl-ribitol phosphate. Strains of *H. influenzae* type b (Hib) cause disease primarily in infants and children <6 years of age. Nontypable strains are primarily mucosal pathogens but occasionally cause invasive disease.

EPIDEMIOLOGY AND TRANSMISSION

H. influenzae, an exclusively human pathogen, is spread by airborne droplets or by direct contact with secretions or fomites. Colonization with nontypable *H. influenzae* is a dynamic process; new strains are acquired and other strains are replaced periodically.

The widespread use of Hib conjugate vaccines in many industrialized countries has resulted in striking decreases in the rate of nasopharyngeal colonization by Hib and in the incidence of Hib infection (Fig. 157-1). Worldwide, invasive Hib disease occurs predominantly in unimmunized children and in those who have not completed the primary immunization series. Most World Health Organization member countries have introduced Hib conjugate vaccination, but a large number of the world's children remain unimmunized, principally in countries without national vaccine programs. Certain groups have a higher incidence of invasive Hib disease than the general population,

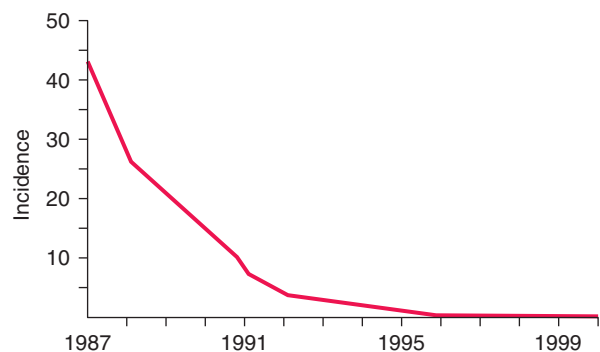


FIGURE 157-1 Estimated incidence (rate per 100,000) of invasive disease due to *Haemophilus influenzae* type b among children <5 years of age: 1987–2000. Fewer than 40 cases per year have been reported since 2000. (Data from the Centers for Disease Control and Prevention.)